DNV·GL

DVR-PP127754
Rev.0

DESIGN VERIFICATION REPORT

Designer/ Manufacturer	: Special and Safety Work (SSW), Riyadh, Saudi Arabia.
DNV GL Project No	: PP127754
Equipment Description	: Blow Out Panel
Codes and Standards	: AISC (9th Edition)

This is to confirm that the structural design of the Blow out Panel (as per drawings listed in Table-1:Design Examination Document) has been reviewed and found to be in compliance with the applicable codes and standards mentioned in this Design Verification Report.

Scope of Verification:

The verification of the design of blow out panel is limited to review of structural analysis submitted in compliance with AISC (9th Edition) as applicable with regrads to local in-place strength against uniformly distributed static equivalent pressure and it is openable under 2 psi.

No other codes / standards have been referred to and it is concluded that client would satisfy themselves that requirements of other relevant codes / standards / national regulations e.g. those mentioned as reference in design calculations submitted etc., are complied with suitably.

DNV GL review is limited to the structure of blow out panel itself.

Table-1: Design Examination Document:

S. No.	Drawing/Document No.	Revision	Drawing/Document Title	Status
1.	SSW-FSC-01	03	Blow Out Panel - Plan, Elevation and Details	Reviewed

Verification Limitations:

Design Parameters:

Parameter	Description
Maximum Pressure Load to be openable	2 psig
Overall Window frame Size	500 mm width and 500mm height

Other design data are considered as per drawing listed in Table-1.

Limitations and Notes:

(1) The design appraisal is limited to the static in place analysis of the steel window frame of Blow Out Panel and any aspects towards adequacy of anchoring arrangement, inspection and testing shall be ensured by the manufacturer.

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(2) The steel frame is found suitable to withstand the maximum pressure load as stated and the concrete part is excluded from our scope.
(3) Welding of the panel shall be full penetration.
(4) Panel structure steel must have yield strength not less than 250 MPa as used in design calculation. All material used for Panel and the hardware components like window frame anchors, etc. including sealants along with fabrication and installation arrangements shall meet the blast rating specifications and structural quality as per the client requirements, If any.
(5) Present appraisal has been carried out with the blow out pressure load indicated in the submitted calculations. Other loads/ load factors like brittle fracture, etc are not considered in this reivew.

Validity Period:

No validity period is stated on the design verification report as it only confirms the conformity with specified regulations / standards at the time of issue. Change in regulations or change in design may have influence on the validity.

Signed for **Det Norske Veritas AS, Abu Dhabi Branch**

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for Rajeev Sharma
Discipline Lead – Static, Container and Lifting Appliance

Balaji Selvaraj
Mechanical Engineer - VMC

OPMAE 646

Date: 08th March 2015
Place: Dubai, UAE